FACULTY OF ENGINEERING AND TECHNOLOGY
CEF 440: INTERNET PROGRAMMING AND MOBILE PROGRAMMING

Names	Matricule
Suh Brandon Njuiebu	FE23A143
Akumo Lucien Ndifor	FE23A010
Formin Rollins Nkengfua	FE23A058
Rosy Winko Nchako	FE23A138
Munongo Bryce Mukete Jnr	FE23A099

Course Instructor: Dr. Nkemeni Valery

TASK 2: Requirement Gathering

Introduction

This report presents a comprehensive analysis of survey data collected from 75 respondents to guide the development of a crowdsensed mobile application for network coverage prediction.

Objectives

As part of our preliminary work for the Crowdsensed Drive Test project, we conducted a targeted survey to understand the hardware and network environment in Buea.

The goal was to collect empirical data on user device types, operating system (OS) versions, and network provider preferences. This information is vital because it determines the technical constraints of the sensing application we are building.

1. Stakeholder identification

a) Primary Stakeholders

These are the people who interact with the system daily.

- **Mobile Subscribers:**
  - **Role:** The "Sensing Nodes."
  - **Interest:** They provide the RSRP/GPS data. They care about privacy, battery life, and having a tool that helps them find better signal areas on campus.
- **The System Developer:**
  - **Role:** Architect and Administrator.
  - **Interest:** You need a functional database and high-quality data to implement your coverage prediction algorithms.
- **Mobile Network Operators e.g., MTN, Orange:**
  - **Role:** The Data Consumers.
  - **Interest:** They benefit from the "Heat Maps" generated by your app to optimize their cell towers without the high cost of manual drive testing.

b) Secondary Stakeholders

These groups don't use the app but are affected by its results.

- **Government Regulators:**
  - **Role:** Quality Control.
  - **Interest:** Using crowdsourced data to verify if MNOs are meeting the minimum signal requirements they promised to the public.
- **Tower Companies:**
  - **Role:** Infrastructure Providers.
  - **Interest:** Knowing where "Coverage Holes" exist to plan the physical construction of new base stations.
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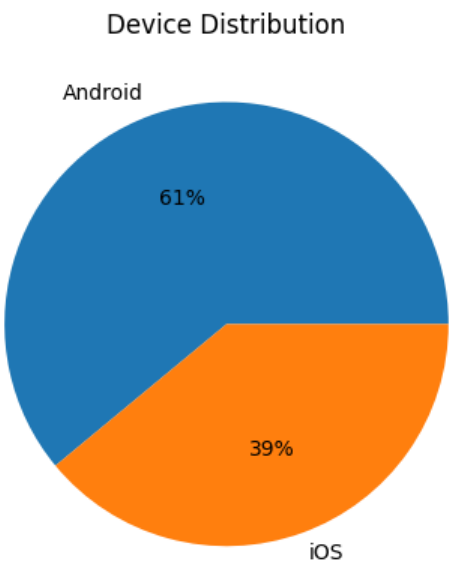
Stakeholder	Role in System	Key Requirement
End Users (Crowd)	Data Generation	Low battery consumption and data privacy.
System Admin	Data Management	Scalable backend and efficient "Store and Forward" logic.
Network Analysts	Data Visualization	High-accuracy GPS tagging and clear "Heat Maps."
MNOs	Implementation Target	Validated Signal-to-Noise Ratio (SINR) metrics.

2. Requirement gathering techniques

Data Analysis & Interpretation

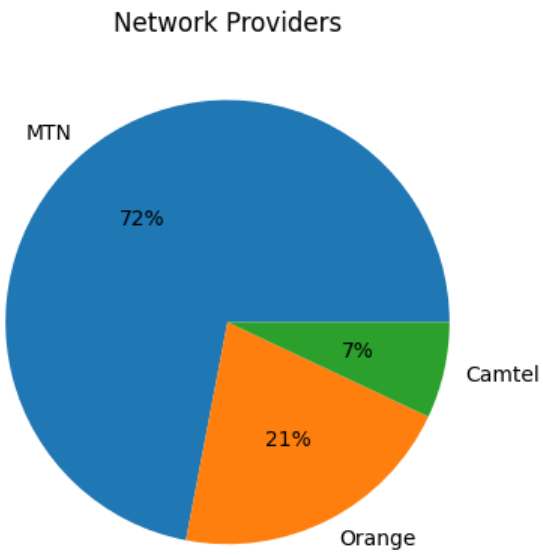
a. Device Distribution

Most users (61%) use Android, indicating primary platform focus.



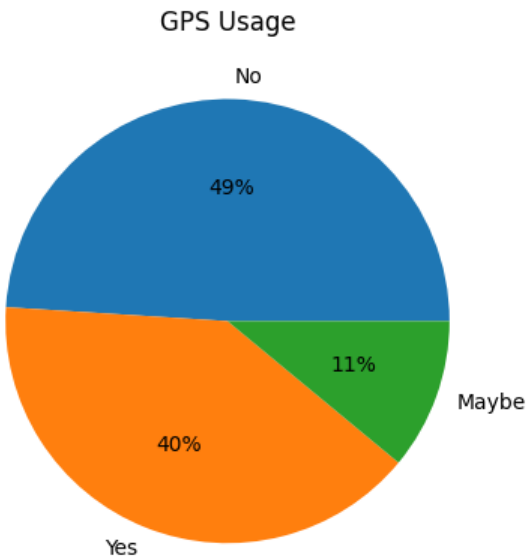
b. Network Providers

MTN dominates usage, suggesting optimization priority.



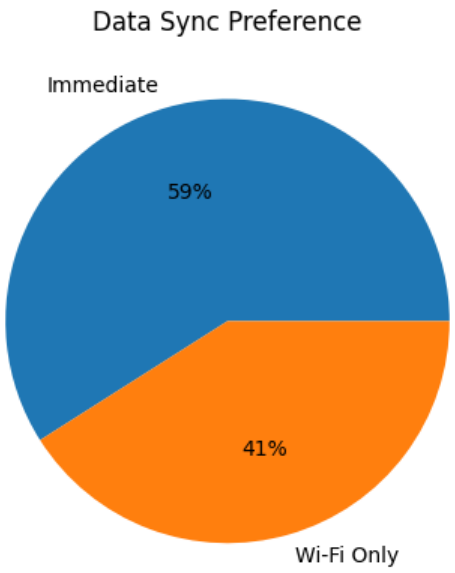
c. GPS Usage

Many users do not enable GPS, requiring efficient permission handling.



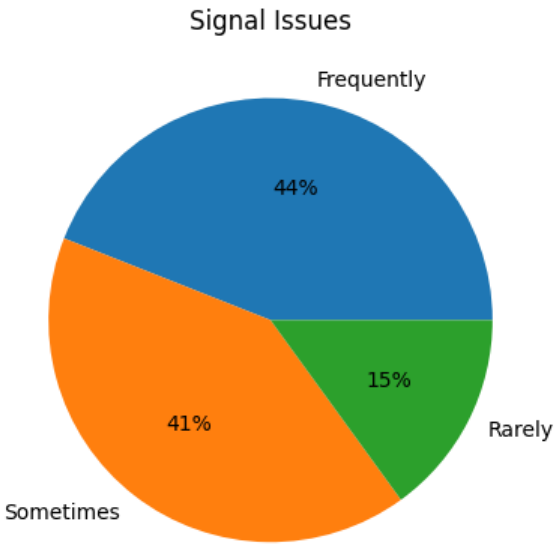
d. Data Sync Preference

Users need flexible sync options.



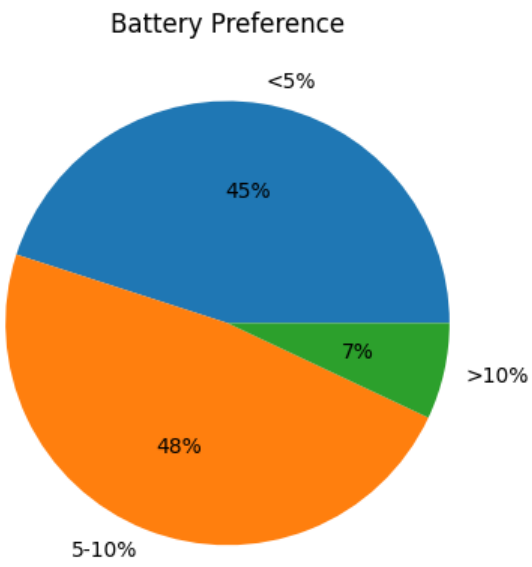
e. Signal Issues

85% experience unreliable connectivity despite signal.



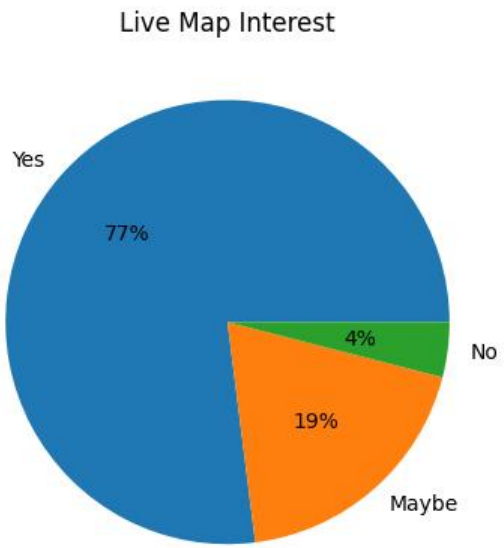
f. Battery Preference

Battery efficiency is critical.



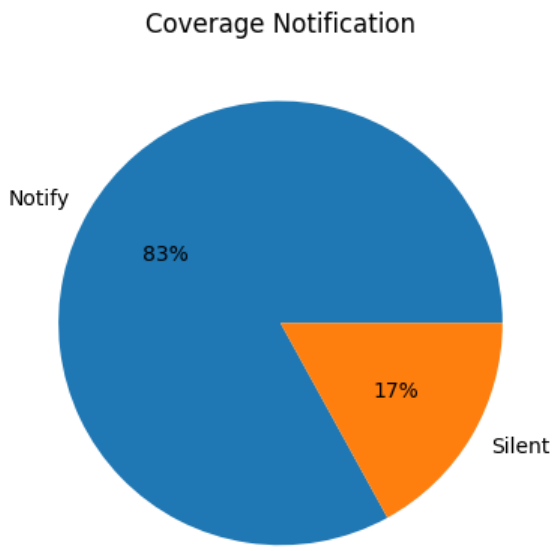
g. Live Map Interest

Strong demand for live signal mapping.



h. Coverage Notification

Users prefer real-time alerts.



I. Key Insights

• A large majority of users experience unreliable connectivity, proving traditional signal indicators are insufficient.
• High interest in live mapping confirms strong demand for real-time coverage visualization.
• Privacy concerns highlight the need for anonymization and transparency.
• Battery constraints require optimized sensing techniques.

II. System Requirements

Functional Requirements:
- Real-time data collection
- Coverage prediction
- Live signal visualization
- User notifications

Non-Functional Requirements:
- Privacy protection
- Low battery usage
- Scalability
- Fast performance

3. Data Gathering

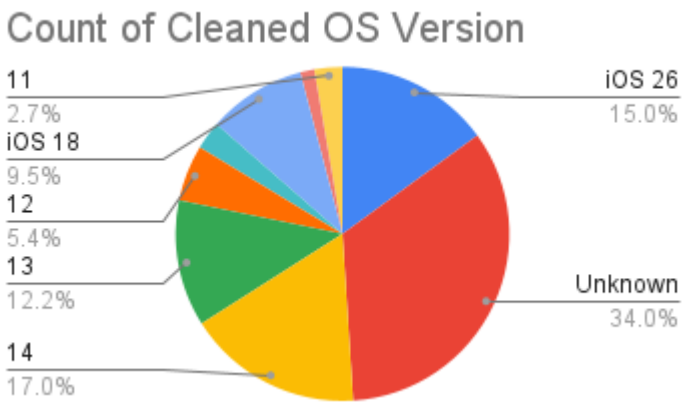
The raw data gathered was initially inconsistent due to manual user input and the inclusion of instructional text from the survey platform. To ensure **data integrity** for our system design, The following cleaning steps was performed:

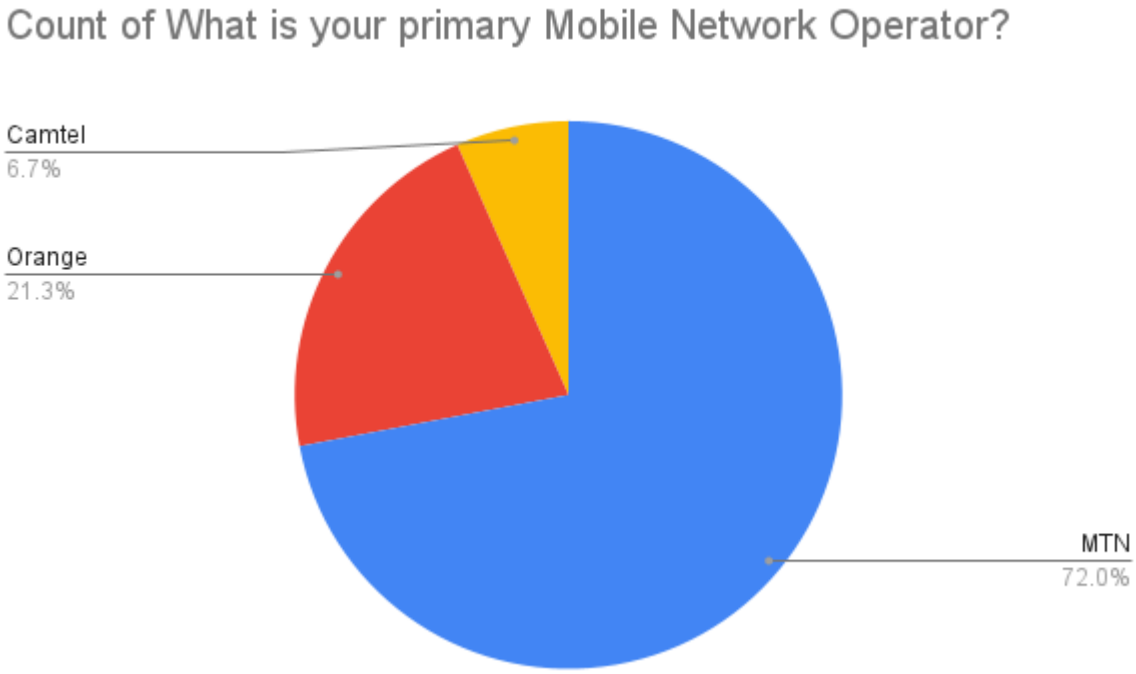
- **Standardization of OS Versions:** Many responses contained parenthetical instructions (e.g., *"iOS 18 (Previous major version)"* or *"iOS 17 or older"*). The strings was stripped to isolate the numerical version. This allows us to accurately calculate which API levels our app must support.
- **Normalization of Network Operators:** User entries for ISPs were inconsistent (e.g., "mtn," "MTN Cameroon," or "MTN"). These were consolidated into four primary categories: **MTN, Orange, Camtel, and Nexttel**.
- **Handling "Unknown" Responses:** A significant number of participants (roughly **30%**) were unable to identify their specific OS version. Instead of discarding this data, I categorized it as **"Unknown."** This high percentage is a critical finding, proving that our app must be able to auto-detect system parameters without user intervention.
- **Validation:** All 75 rows were reviewed for duplicate or "test" entries. The final dataset consists of 75 unique, valid data points.

a. Analysis and Link to Project Goals

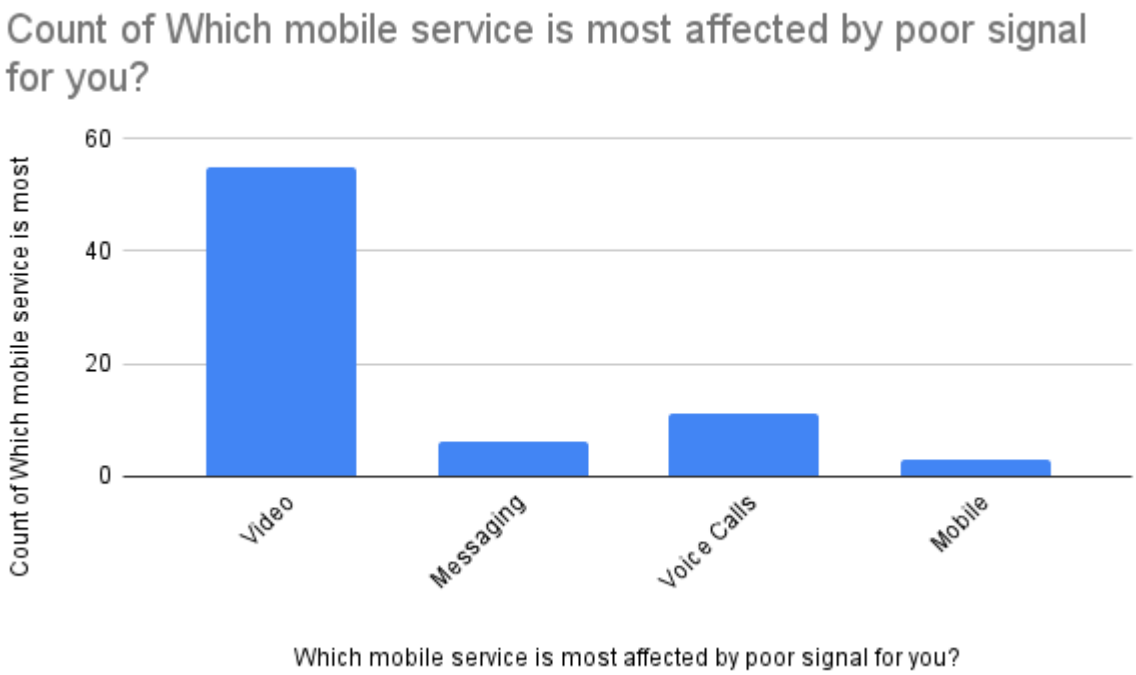
The cleaned data provides the technical foundation for our next phase. The following links between the data and the project goals were identified:

- **Platform Prioritization:** With **61.3%** of users on Android (majority on Android 14), our sensing module must prioritize the **Android TelephonyManager API**. However, the spread across versions 11–14 means we must implement backward-compatible code.

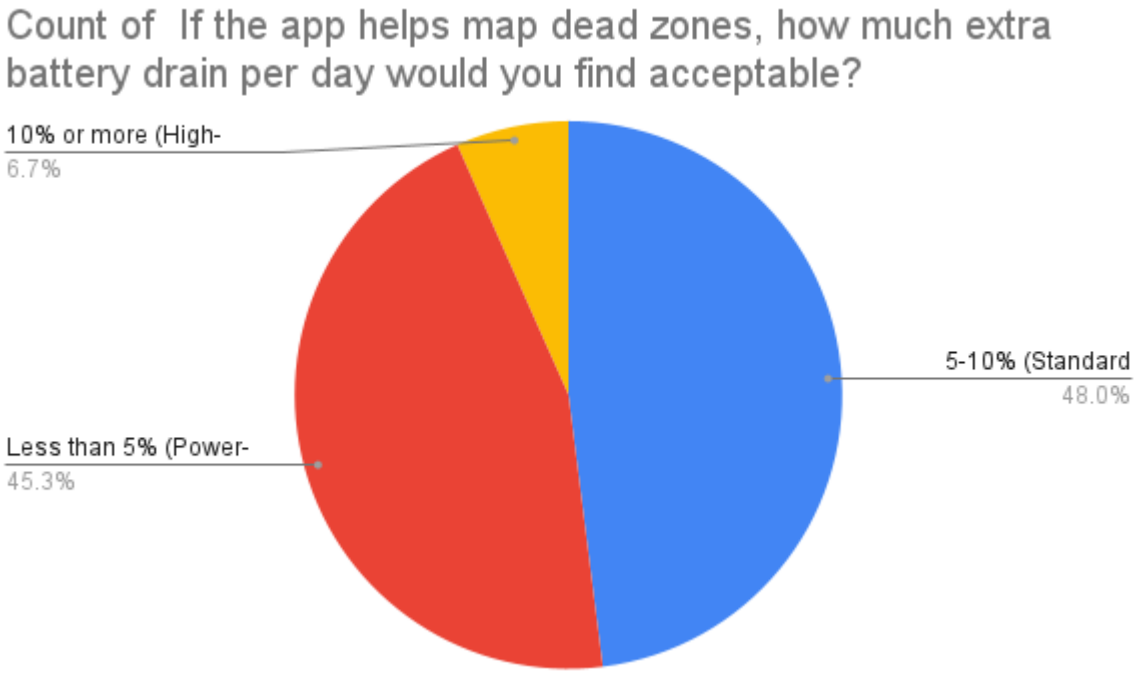




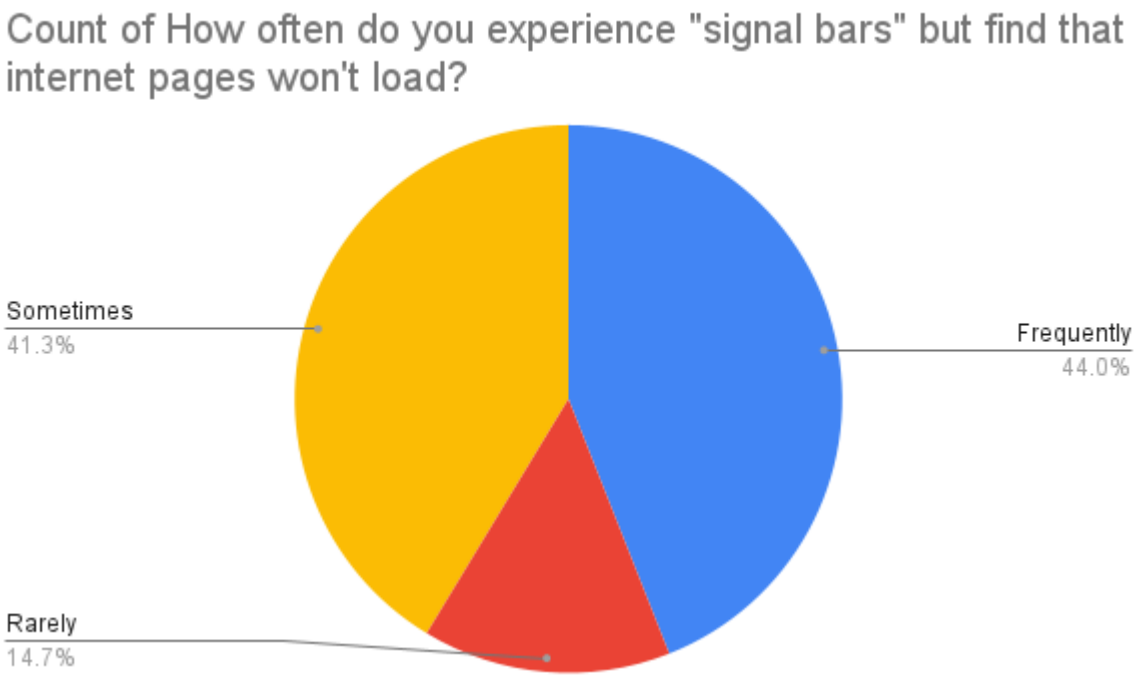
- **Network Performance Justification: 73.3%** of users reported that **Video Streaming** is most affected by poor signal. This justifies our project's focus on measuring not just "bars" (RSSI), but actual signal quality (RSRQ) and throughput.



- **System Constraints (Battery & Sync): 48%** of users will only tolerate less than **5% battery drain**. This is a major constraint; our "drive test" sensing cannot be constant and must use optimized polling intervals to save power.



- **The Ghost Signal Phenomenon:** Over **40%** of participants reported having signal bars but no internet. This finding validates our Topic 3 objective: we are not just mapping coverage; we are mapping **performance**.



4. Cleaning Report

Data cleaning is a measure taken to make sure the information we have is clear, concise, and relevant for using.

Tools used for data cleaning:

- Google Sheets
- Microsost Excel
- Google Forms

To clean our data we exported all our data from google form into excel sheets and we performed the following data cleaning techniques

- **Standardization:** This involves forcing inconsistent entries into a single, uniform format.
- **Deduplication:** Identifying and removing identical records that were submitted more than once.
- **Logical Consistency Checking:** Comparing two different fields to see if they contradict each other.
- **Handling Missing Values:** Deciding what to do when a data point is incomplete and either imputing data or deleting the row.
- **Outlier Detection & Range Filtering (Noise Removal):** This technique removes data points that are physically impossible or mathematically illogical.

5. User Reluctance Assessment

A process to identify psychological, practical, and technical barriers that stop users from downloading, keeping, or using an app.

Reluctance types:

- **Privacy/Permission:** Users deny location, contacts, etc. Fix: explain why you need access.
- **Complexity:** Users quit long onboarding. Fix: keep it under 3 taps, offer guest mode.
- **Performance:** Slow loads or crashes cause uninstalls. Fix: target <1% crash rate, optimize assets.
- **Value:** No clear reason to return. Fix: demonstrate core benefit immediately.
- **Trust**: Low ratings or unknown developer. Fix: respond to reviews, show social proof.

- **Resource:** App too large or drains battery. Fix: offer a lite version, limit background activity.
- **Key metrics:** Permission grant rate, onboarding completion, crash rate, D1/D7/D30 retention, store ratings, and battery usage.
- **Simple workflow:** Interview before launch → track drops in beta → survey users live → alert on threshold breaches.
- **Bottom line:** Users leave due to many small frictions, not one big problem. Fix the highest-impact barrier first, then re-measure.

5. Conclusion

The findings validate the need for a crowdsensed application capable of predicting network performance accurately. User expectations emphasize accuracy. The cleaning process has successfully transformed raw user feedback into a structured technical baseline. We have identified that the app must be **Android first**, **highly automated** (due to user technical gaps), and **power-efficient**.